

WASP-39b

R-1033

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_260326 · created 2026-07-30T06:29:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461125.799 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1848 +/- 0.0091
Transit depth (Rp/Rs)^2	3.41 +/- 0.34 [%]
Orbital Inclination (inc)	87.58 +/- 0.78
Airmass coefficient 1 (a1)	1.182 +/- 0.017
Airmass coefficient 2 (a2)	-0.1242 +/- 0.0032
Scatter in the residuals of the lightcurve fit is	1.37 %
Best Comparison Star	None
Optimal Aperture	4.34
Optimal Annulus	23.42
Transit Duration (day)	0.1213 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1848 $\pm$ 0.0091 vs 0.1457 (deviation 4.3 σ)
Duration fit vs published	0.1213 d vs 0.12 d (deviation 0.15 σ)
Mid-time O-C	-0.4 min
Depth SNR	20.3
Residual scatter	1.37 %

Light curve

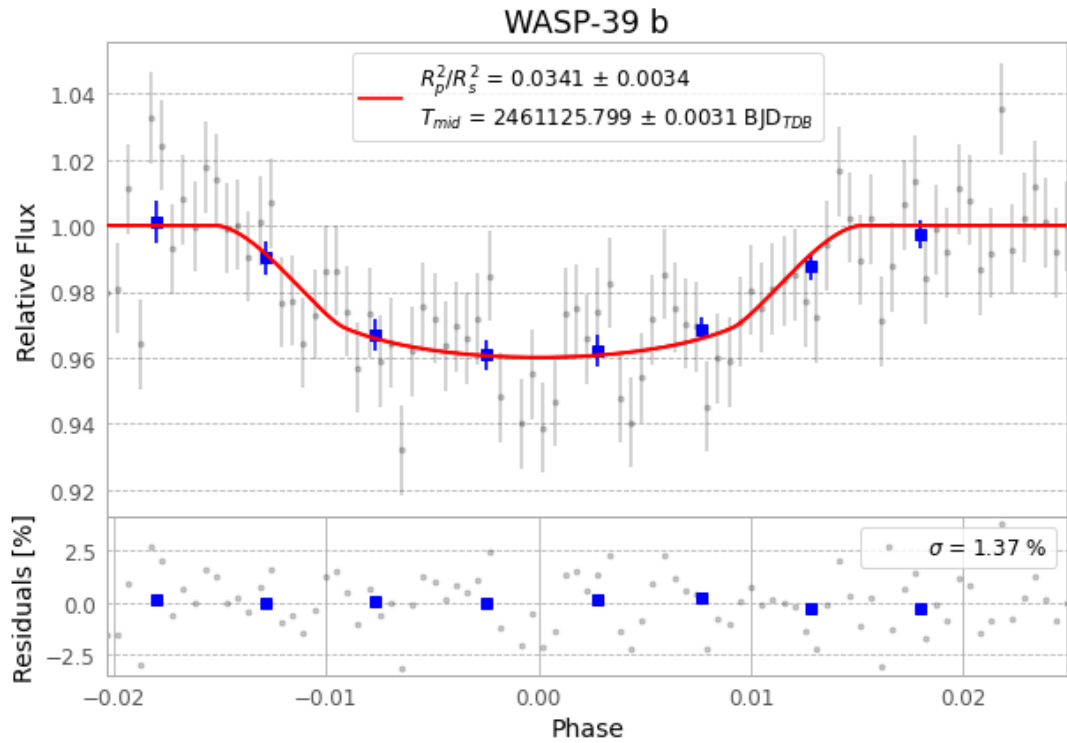


Figure 1 — FinalLightCurve_WASP-39 b_2026-03-25.png (SHA-256 0be1a89f947cf226...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 34c1cf400a34da5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9131c47

Data provenance

89 FITS frames (58 MB), WASP-39260326050315.FITS → WASP-39260326092715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-260326.log (6dd226adb38b...), FinalLightCurve_WASP-39 b_2026-03-25.png (0be1a89f947c...), FinalLightCurve_WASP-39 b_2026-03-25.csv (2ae9093e416b...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:29Z.